

Witterung und Klima in Mitteleuropa. By Dr. Hermann Flohn. Stuttgart (S. Hirzel), 2nd Edition 1954. Pp. 214; 33 Figs., tables, plates. 24 DM.

The first edition of this distinctive work under a slightly different title (*Deutschland* instead of *Mitteleuropa*) first appeared in 1942 and may not have received wide notice outside Germany. Reset in roman type with interesting additional chapters on Austria by Dr. F. Lauscher and on Switzerland by Dr. M. Schüepp and an extended bibliography, the book retains its unique character as a discussion of the mechanism of a regional climate in terms familiar to the synoptic meteorologist. This aspect of climatology, which has in relatively recent years become known as synoptic or dynamical climatology, is termed by Professor Flohn 'modern climatology' while the more established studies of the climatic elements separately, which give special prominence to normal values, are termed 'classical climatology.' The terminology will not be readily accepted by everyone for the words 'modern' and 'classical' in no way suggest the different methods of treatment both of which are necessary and useful and will continue to be so. Indeed for most practical purposes of applied climatology the 'classical' method remains the more valuable although it is scientifically somewhat arid. No doubt the climatological study of synoptic behaviour, weather types, air-mass frequencies, frontal passages and the like is important for an understanding of climate and is particularly useful as background knowledge for the synoptic weather forecaster but it does not provide ready answers to most of the questions put to the climatologist by his public.

The reader who is interested in the usual climatological statistics will not therefore find the information in this book which has very few tables and no climatic map. But these things can readily be found elsewhere and there is no doubt that this discussion of the structure of climate is a valuable supplement. An interest in the climate of central Europe is indeed hardly necessary to make Professor Flohn's book worth acquiring, for he is as much concerned with ideas and methods of general meteorological interest as with facts peculiar to central Europe.

The subject matter does not lend itself readily to formal text-book presentation and the different chapters are rather in the nature of separate essays. The style tends to be discursive with numerous digressions and the 184 pages of text, almost unrelieved by tables or diagrams, are likely to present difficulty to all but the fluent reader of German. The continual reference to original papers is also a distraction and one may wonder whether a bibliography running to some twenty odd pages with over 500 entries would not have been more useful had it been more selective. There are no fewer than 191 references for the study of 'singularities,' presenting a formidable prospect to the student who may wish to pursue this subject.

The first chapter is of an introductory and historical nature stating the author's attitude to climatology. The remainder deal amongst other things with weather types, air masses, large-scale weather types (*Grosswetterlagen*) and climatic oscillations, but perhaps the most striking chapter is that in which the author takes us on a conducted tour through the year pointing out the characteristics of the seasons with special attention to singularities and regularities. It is a difficult art to make a coherent story from the synoptic weather of the year in the zone of the 'turbulent baroclinic westerlies' and it requires almost an act of faith to recognize as many as seven 'waves' to the early summer 'monsoon.' It is unlikely that the sceptic will be persuaded to accept this analysis from the evidence provided and until the phenomena have been more strictly defined and statistically tested there will remain those who regard most 'singularities' as chance coincidences in the past with no prediction value for the future. But the subject has received relatively slight attention outside Germany and is much too important to be neglected.

Mention must also be made of the excellent diagrams collected at the end of the volume, especially those giving the long-term mean values or frequencies for each day of the year of days with föhn, air-mass change, cloudy and sunny days, sunshine duration, precipitation, thunderstorms, pressure and temperature at selected stations. One may question the validity of identifying the many peaks and troughs of a few days' duration with entities of climatological significance, but the facts are striking and it should be the aim of 'modern climatology' to give them a satisfying objective explanation.

R.C.S.

On the origin of the solar system. By H. Alfvén. International series of monographs on physics, Oxford (Clarendon Press), 1954. Pp. viii, 194; 47 Figs., 15 tables. 30s.

In this book a novel theory of the origin of the solar system is discussed. It is a theory that does not require the intervention of other celestial objects for generating the sun, planets, and satellites, but merely the interaction of the sun with infalling gas and dust clouds. The author, who has played a considerable part in impressing the importance of magnetic interactions on astronomical thought, makes these play a large part here also; and he is no doubt on strong ground there.

The tracing of the main processes of the theory is, however, not convincing. According to his theory, the materials falling towards the sun have to be suspended against gravity for very long periods of time in order that diffusive separation of chemicals may take place. A magnetic field is postulated as causing such suspension, but a field can only do this if it is firstly strong enough and secondly if its lines of force do not link the gravitating body. The required strength is 10^6 times greater than the estimates of the present solar magnetic field, and this assumption is discussed. But the completely vital question of linkage is not discussed, and the field is treated as a solar dipole. A reference is made to this basic conflict but no resolution is suggested.

The well-known cosmogonical theory of the distribution of the angular momentum within the solar system is also treated only sketchily. No detailed mechanism of the transfer of this from the sun to a surrounding gas cloud is given, and one may wonder whether the assumption of a strong solar magnetic field is really all that is required.

Despite the severe criticism that can be raised against the theory as a whole, there is no doubt that the book contains stimulating discussions of various points of magnetodynamics which may well be of great importance in a future theory.

The index is somewhat inadequate; for example, of the seven authors mentioned on p. 1, only four appear in the author index. The quality of the printing, the plates, and figures is of the usual high standard of the Oxford University Press.

Thomas Gold

Dynamik der Zyklonen. By P. Raethjen. Leipzig (Akademische Verlag., Geest and Portig), 1953. Pp. xii, 384; 92 Figs., 3 tables. 35 DM.

This is not so much a text-book as an up-to-date report on researches made in all schools of meteorology. It covers a wider field than a narrow interpretation of its title would imply, perhaps inevitably, since the dynamics of cyclones could hardly be dealt with adequately without bringing in considerations of the general circulation. In a critical historical review the author gives an exposition of many early ideas which appear in proper perspective against the background of all the work done and he extracts from earlier work ideas which still appear valid in the light of more recent and comprehensive observations. In this way he bridges the gap between earlier and more recent theories but any forecaster who looks to find a single model of a depression emerging as a building erected on solid foundations will be disappointed. No such model exists but numerous possible causes of depressions are clarified.

The book is divided into two parts, each with seven chapters, the first being descriptive, and the second mathematical. The first part is not only descriptive, but attempts to give the physical ideas underlying the mathematical treatment later in the book. The first two chapters deal with the old basic conceptions in a somewhat critical manner, regarding polar front waves not as boundary waves but as disturbances of a wide-spread state of unstable equilibrium. Only partially then is the wave theory of cyclone development retained. In the third chapter the development of frontal zones is discussed in the light of the accumulating mass of aerological observations and the interaction between depressions and the mean field of motion is clearly described as well as the 'jet-stream,' 'cut-offs' and 'meandering.' The general circulation is discussed in chap. 4, a meridional cross-section demonstrating a type of circulation which will not be acceptable to all.

In chap. 5 the chief possibilities of development, from vorticity considerations, are discussed, first among those being the maintenance of absolute vorticity in meridionally moving air masses. In the remaining two chapters of the first part of the book different models of depressions are discussed and some forecasters may be surprised by the suggestion that high-level depressions may be formed by sinking of the polar stratosphere. Other suggestions in these chapters appear to be debatable.

In the mathematical half of the book the author plunges straight into vorticity and consideration of its variations, while in the following chapter the importance, for the latter, of horizontal divergence and vertical velocity are dealt with. Chaps. 10 and 11 include discussions of the pressure changes due to the vorticity changes and these chapters on vorticity have a disarming simplicity about them. A discussion of large-scale turbulence is given in chap. 12, in relation to frontal zones, so as to explain the development of centres of action, and in chap. 13 the long waves in the zonal circulation (after Rossby) are discussed.

The closing chapter deals with the convectional depression with particular reference to tropical hurricanes.

Some of the ideas of Professor Raethjen will appear strange and perhaps bizarre to British synoptic meteorologists but for those in search of a new approach to the problems of synoptic meteorology this may be intriguing if not stimulating. In these days when synoptic ideas have to be shaped to suit machines there may be a danger of machine-made ideas. There is a novelty about some of the things which this author writes which might, even indirectly, avert this danger.

The book is beautifully printed, though the binding may not suit British taste, and there is a bibliography comprising 840 items.

J.S.F.

Seismology. By K. E. Bullen. London (Methuen), 1954. Pp. viii, 132; 15 diagrams. 8s. 6d.

Seismology is nowadays a vast subject, and the literature is still extending rapidly. To give an adequate account of the technique, the theory and the inferences that can be drawn from earthquake studies requires the nicest discrimination, as well as an author completely *au fait* with the latest developments. Prof. Bullen, whose researches have contributed so largely to our present knowledge of the earth's interior, is exceptionally well qualified for this task, and this new book is a marvel of compression, yet nowhere has clarity been sacrificed in the interests of brevity.

The treatment is quantitative so far as that is reasonable and practicable, and some saving of space is effected by omitting detailed mathematics and giving appropriate references to Prof. Bullen's standard treatise on theoretical seismology. The non-mathematical reader, however, need not be put off; he will